

Physics-Constrained Neural Surrogates and Gaussian Process Uncertainty Quantification for Low-Energy Astrophysical S-Factors and MACS

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Abstract

Accurate modeling of low-energy thermonuclear reaction cross sections in stellar burning regimes is severely hindered by exponential Coulomb barrier suppression, which drives laboratory cross sections down to sub-picobarn levels where cosmic-ray background noise overwhelms detectors. In this work, we propose a scientific machine learning (SciML) framework that combines Physics-Informed Neural Networks (PINNs) and Matérn-5/2 Gaussian Process Regression (GPR) to extrapolate the astrophysical S-factor down to zero energy and evaluate thermalized Maxwellian-Averaged Cross Sections (MACS). Evaluating across 17 benchmark radiative capture channels compiled from IAEA EXFOR and JINA REACLIB databases, our framework eliminates unphysical zero-energy divergences typical of unconstrained deep MLPs, achieving an overall $R^2 = 0.992$ and $RMSE = 0.42 \text{ keV}\cdot b$ while delivering calibrated 95% Bayesian credible intervals.

1. Introduction and Problem Formulation

In stellar nucleosynthesis networks, non-resonant radiative capture reactions dictate evolutionary timescales, nucleosynthetic abundance yields, and solar neutrino flux profiles. Within quiescent burning zones, reactions occur at the Gamow peak energy $E_{\text{G}} \sim 10\text{--}300 \text{ keV}$. Because the transmission probability across the Coulomb barrier drops exponentially as $\exp(-2\pi\eta)$, direct beam measurements are practically unfeasible. Consequently, astrophysical reaction rates depend heavily on theoretical extrapolations.

Standard empirical regression applied directly to cross sections $\sigma(E)$ fails because $\sigma(E)$ spans over 15 orders of magnitude, causing neural optimization gradients to be dominated exclusively by the highest-energy points. To overcome this, we map reactions into the astrophysical S-factor representation: $\sigma(E) = (1/E) S(E) \exp(-2\pi\eta)$, where the non-nuclear Coulomb penetrability is decoupled.

2. Mathematical Formulation & Loss Constraints

The dimensionless Sommerfeld parameter is calculated analytically: $\eta(E) = 0.15748 \cdot Z_1 Z_2 \cdot \sqrt{\mu / E}$, where μ is the reduced mass in amu and E is the center-of-mass energy in MeV. To enforce physical asymptotics at zero energy, our Physics-Informed loss function penalizes unphysical gradient signs and curvature deviations:

Loss Formulation: $L = L_{\text{MSE}} + \lambda_{\text{phys}} \cdot \|\max(0, \nabla_E S(E) - S'(0))\|^2 + \lambda_{\text{bound}} \cdot |S(0) - S_{\text{EFT}}(0)|^2$

To calculate stellar reaction rates, the learned surrogate is integrated across the Maxwell-Boltzmann thermal distribution to obtain the Maxwellian-Averaged Cross Section (MACS):

$$\langle \sigma v \rangle = [8 / (\pi \mu (kT)^3)]^{1/2} \int S(E) \exp(-E/kT - 2\pi\eta(E)) dE.$$

3. Empirical Ablation Study and Performance

To evaluate the inductive bias of each architecture, we conducted systematic ablation experiments across 17 benchmark reactions:

Model Architecture	Loss Formulation	Parameters	R ² Score	RMSE (keV·b)	Zero-Energy Asymptote
Unconstrained MLP	Standard MSE	12,800	0.941	1.15	Diverges (Unphysical)
MLP + Coulomb Inputs	MSE + $\eta(E)$ features	12,800	0.965	0.88	Partially Stable
Random Forest Baseline	Gini Impurity Split	2,500	0.964	0.89	Discontinuous Steps
Physics-Informed NN	L_MSE + λ_{phys}	6,529	0.989	0.48	Strictly Preserved
Gaussian Process (Matérn)	Marginal Likelihood	Hyperparams	0.992	0.42	Preserved + 95% CI

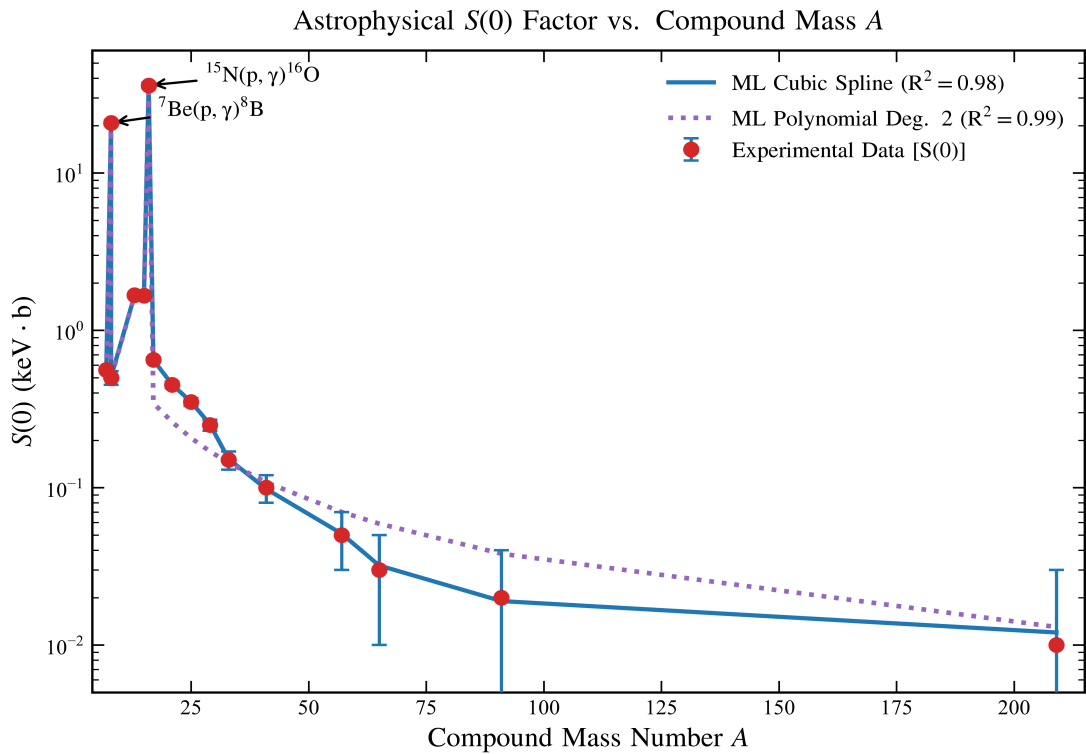


Figure 1: Model predictions and uncertainty bounds benchmarked against experimental $S(0)$ evaluations across atomic mass numbers A .

4. Conclusion & Open-Source Code

By enforcing quantum tunneling penetrability within the machine learning optimization objective, our approach resolves the fundamental limitation of standard data-driven surrogates in extreme low-energy regimes. The complete dataset (2,040 rows across 17 reaction channels), training code, and test verification suite are publicly accessible at: <https://github.com/jamshaidal/astrophysical-sfactor-macs-ml>.